

# Online Appendix

## Bitter Pills to Swallow: The Enforcement Costs of Health Litigation

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*Not for print — available as supplementary material.*

### Online Appendix

This appendix provides tables and figures from the main text, robustness checks, and supplementary evidence.

#### *A.0 Supporting Descriptives, Balance, and Secondary Outcome Tables*

Section A.1 examines the sensitivity of the main estimates to the choice of winsorization level. Section A.2 assesses the robustness of the “under the gun” estimates to progressive inclusion of control variables. Section A.3 presents distributional evidence on key outcome variables and additional descriptive figures. Section A.4 presents geographic descriptive evidence on administrative demand, complementing the litigation maps shown in Section ???. Section A.5 presents an event study around the first court order for each item. Section A.6 examines heterogeneity in the urgency premium along several dimensions.

#### *A.1 Winsorization Sensitivity*

Our baseline estimates use 1%/99% winsorization to limit the influence of extreme outliers while preserving meaningful variation in the data. To verify that our results are not driven by this specific choice, Tables [A.8–A.11](#)

Table A.1: Descriptive Statistics by Purchase Type

|                                        | Ordinary  |              | Administrative |              | Litigated |             | Diff                    | Diff                     |
|----------------------------------------|-----------|--------------|----------------|--------------|-----------|-------------|-------------------------|--------------------------|
|                                        | Mean      | (SD)         | Mean           | (SD)         | Mean      | (SD)        | (O-L)                   | (A-L)                    |
| <i>Panel A: Levels</i>                 |           |              |                |              |           |             |                         |                          |
| Reference Price                        | 909.17    | (5,959.41)   | 226.38         | (2,257.35)   | 370.39    | (2,540.20)  | 538.78***<br>[0.000]    | -144.01***<br>[0.000]    |
| Negotiated Price                       | 477.51    | (2,915.88)   | 148.03         | (1,177.52)   | 260.25    | (1,561.81)  | 217.26***<br>[0.000]    | -112.22***<br>[0.000]    |
| Quantity                               | 31,487.15 | (163,507.29) | 117,146.00     | (356,962.37) | 9,646.19  | (82,683.10) | 21,840.96***<br>[0.000] | 107,499.81***<br>[0.000] |
| N. Bidding Firms                       | 3.17      | (1.93)       | 2.43           | (1.35)       | 2.20      | (1.19)      | 0.97***<br>[0.000]      | 0.23***<br>[0.000]       |
| <i>Panel B: Log Transformations</i>    |           |              |                |              |           |             |                         |                          |
| Log Reference Price                    | 0.926     | (2.723)      | 0.909          | (2.541)      | 1.557     | (2.446)     | -0.632***<br>[0.000]    | -0.648***<br>[0.000]     |
| Log Negotiated Price                   | 0.319     | (2.862)      | 0.480          | (2.604)      | 1.024     | (2.568)     | -0.705***<br>[0.000]    | -0.544***<br>[0.000]     |
| Log Quantity                           | 6.947     | (2.678)      | 7.472          | (3.160)      | 6.087     | (2.083)     | 0.860***<br>[0.000]     | 1.385***<br>[0.000]      |
| Log N. Firms                           | 0.984     | (0.590)      | 0.747          | (0.529)      | 0.665     | (0.493)     | 0.319***<br>[0.000]     | 0.081***<br>[0.000]      |
| <i>Panel C: Tender Characteristics</i> |           |              |                |              |           |             |                         |                          |
| Successful Tender (%)                  | 0.859     | (0.348)      | 0.869          | (0.338)      | 0.908     | (0.288)     | -0.050***<br>[0.000]    | -0.040***<br>[0.000]     |
| Observations                           | 136,250   |              | 15,371         |              | 45,351    |             |                         |                          |

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch *t*-tests; *p*-values in brackets. \*\*\* *p*<0.01, \*\* *p*<0.05, \* *p*<0.1.

replicate the main regression tables under three alternative winsorization treatments: no winsorization (Panel A), the 1%/99% baseline (Panel B), and a more aggressive 5%/95% winsorization (Panel C).

Table A.8 reports the sensitivity of the reference price estimates. The positive coefficient on urgent purchases is stable across all three winsorization levels: the preferred specification (column 3) yields estimates of 0.029 (no winsorization), 0.027 (baseline), and a similar magnitude with aggressive winsorization. The qualitative pattern—a large raw premium that attenuates substantially with PBU fixed effects—is preserved throughout, indicating that the reference price result is not driven by outliers.

Table A.9 presents the corresponding sensitivity analysis for negotiated prices. The total effect of urgency on negotiated prices is positive and statistically significant across all winsorization levels. The coefficient magnitudes are slightly larger without winsorization (reflecting the influence of extreme

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

|                                          | Administrative |              | Litigated |             | Diff          | p-value |
|------------------------------------------|----------------|--------------|-----------|-------------|---------------|---------|
|                                          | Mean           | (SD)         | Mean      | (SD)        | (A-L)         |         |
| <i>Panel A: Procurement Outcomes</i>     |                |              |           |             |               |         |
| Reference Price                          | 226.38         | (2,257.35)   | 232.56    | (1,432.18)  | -6.18         | [0.752] |
| Negotiated Price                         | 148.03         | (1,177.52)   | 174.79    | (1,052.56)  | -26.75**      | [0.013] |
| Quantity                                 | 117,146.00     | (356,962.37) | 9,306.87  | (80,252.25) | 107,839.13*** | [0.000] |
| Log Reference Price                      | 0.91           | (2.54)       | 1.43      | (2.32)      | -0.52***      | [0.000] |
| Log Negotiated Price                     | 0.48           | (2.60)       | 0.89      | (2.44)      | -0.41***      | [0.000] |
| Log Quantity                             | 7.47           | (3.16)       | 6.18      | (1.99)      | 1.29***       | [0.000] |
| <i>Panel B: Market Structure</i>         |                |              |           |             |               |         |
| N. Bidding Firms                         | 2.427          | (1.351)      | 2.155     | (1.120)     | 0.272***      | [0.000] |
| Log N. Firms                             | 0.747          | (0.529)      | 0.650     | (0.483)     | 0.097***      | [0.000] |
| Successful Tender (%)                    | 0.869          | (0.338)      | 0.914     | (0.280)     | -0.045***     | [0.000] |
| <i>Panel C: Purchase Characteristics</i> |                |              |           |             |               |         |
| Electronic Auction                       | 0.985          | (0.123)      | 0.941     | (0.236)     | 0.044***      | [0.000] |
| Observations                             | 15,371         |              | 41,491    |             |               |         |

*Notes:* Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%.  $p$ -values from Welch  $t$ -tests in brackets. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

values) and slightly smaller with aggressive winsorization, but the economic message remains unchanged: urgent purchases are associated with higher negotiated prices, with the preferred estimate ranging from approximately 3% to 7%.

Table A.10 examines the robustness of the bidder participation results. The negative effect of urgency on the number of bidding firms is remarkably stable across winsorization levels, with preferred estimates in the range of  $-0.05$  to  $-0.06$  log points. This stability is expected, since the number of firms is a count variable with limited extreme values, making it less sensitive to winsorization.

Table A.11 replicates the tender success analysis. The positive coefficient on urgency—indicating that urgent tenders are more likely to succeed—is

Table A.3: Reference Prices

|                       | (1)                 | (2)                 | (3)               | (4)               |
|-----------------------|---------------------|---------------------|-------------------|-------------------|
| Urgent Purchase       | 0.164***<br>(0.056) | 0.156***<br>(0.050) | 0.027*<br>(0.014) | 0.027*<br>(0.014) |
| Item FE               | Yes                 | Yes                 | Yes               | Yes               |
| Year FE               | No                  | Yes                 | Yes               | No                |
| Year-Month FE         | No                  | No                  | No                | Yes               |
| PBU FE                | No                  | No                  | Yes               | Yes               |
| Observations          | 196,899             | 196,899             | 196,896           | 196,896           |
| Within R <sup>2</sup> | 0.003               | 0.003               | 0.000             | 0.000             |

*Notes:* Dependent variable: log reference price. Sample: winners only, items with both ordinary and litigated purchases. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

robust across all winsorization levels. The magnitude is consistent at approximately 2 percentage points in the preferred specification, reinforcing the interpretation that procurement officials under pressure accept less favorable terms to ensure compliance.

#### A.2 Under the Gun: Progressive Controls

Table A.12 examines the sensitivity of the “under the gun” estimate to the progressive inclusion of control variables and to the choice of winsorization level. Starting from the baseline specification with item + year + PBU fixed effects (column 1), we sequentially add log quantity (column 2), log reference price (column 3), log number of firms (column 4), and finally replace year FE with year-month FE (column 5). Each row within a panel shows how the coefficient on the administrative indicator responds to additional controls.

The key finding is that the administrative indicator remains negative across all specifications and winsorization levels, indicating that litigated purchases are consistently more expensive than administrative ones. The coefficient is largest and most precisely estimated in the baseline specification

Table A.4: Quantities

|                       | (1)               | (2)               | (3)               | (4)               |
|-----------------------|-------------------|-------------------|-------------------|-------------------|
| Urgent Purchase       | -0.264<br>(0.183) | -0.279<br>(0.181) | -0.058<br>(0.056) | -0.065<br>(0.058) |
| Item FE               | Yes               | Yes               | Yes               | Yes               |
| Year FE               | No                | Yes               | Yes               | No                |
| Year-Month FE         | No                | No                | No                | Yes               |
| PBU FE                | No                | No                | Yes               | Yes               |
| Observations          | 196,899           | 196,899           | 196,896           | 196,896           |
| Within R <sup>2</sup> | 0.003             | 0.003             | 0.000             | 0.000             |

*Notes:* Dependent variable: log quantity. Sample: winners only. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

without additional controls (column 1), where it captures the total effect including indirect channels through quantity and competition. As controls are added, the coefficient attenuates—consistent with some of the price premium operating through the quantity and competition channels—but the sign is preserved throughout. This pattern confirms that the sanction channel has an economically meaningful direct effect on prices beyond its indirect effects through procurement conditions.

### A.3 Distributional Evidence

The regression estimates in the main text capture average treatment effects. The figures below complement these estimates by showing how the *distributions* of key outcome variables differ across purchase types, providing visual evidence of the systematic shifts that underlie our regression results.

Figure A.3 plots kernel density estimates of log reference prices separately for ordinary, administrative, and litigated purchases. The rightward shift of the litigated distribution relative to ordinary purchases is clearly visible, consistent with the positive coefficient on urgency in Table A.3. The administrative distribution lies between the other two, reflecting the intermediate

Table A.5: Participant Firms

|                               | (1)                  | (2)                  | (3)                  | (4)                  |
|-------------------------------|----------------------|----------------------|----------------------|----------------------|
| <i>Panel A: Total Effect</i>  |                      |                      |                      |                      |
| Urgent Purchase               | -0.101***<br>(0.023) | -0.113***<br>(0.022) | -0.056***<br>(0.014) | -0.054***<br>(0.015) |
| <i>Panel B: Direct Effect</i> |                      |                      |                      |                      |
| Urgent Purchase               | -0.091***<br>(0.014) | -0.102***<br>(0.011) | -0.042***<br>(0.012) | -0.040***<br>(0.013) |
| Log Quantity                  | 0.063***<br>(0.004)  | 0.064***<br>(0.004)  | 0.065***<br>(0.003)  | 0.065***<br>(0.003)  |
| Item FE                       | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE                       | No                   | Yes                  | Yes                  | No                   |
| Year-Month FE                 | No                   | No                   | No                   | Yes                  |
| PBU FE                        | No                   | No                   | Yes                  | Yes                  |
| Obs. (Panel A)                | 81,805               | 81,805               | 81,801               | 81,801               |
| Obs. (Panel B)                | 81,793               | 81,793               | 81,789               | 81,789               |
| Within R <sup>2</sup> (A)     | 0.004                | 0.006                | 0.001                | 0.001                |
| Within R <sup>2</sup> (B)     | 0.063                | 0.069                | 0.043                | 0.044                |

*Notes:* Dependent variable: log number of bidding firms. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

planning conditions of this purchase type.

Figure A.4 presents the analogous comparison for log negotiated prices. The distributional shift is qualitatively similar to reference prices, with litigated purchases showing higher negotiated prices on average. Notably, the litigated distribution also exhibits a somewhat thicker right tail, suggesting that a subset of litigated purchases involves particularly large price premiums.

Figure A.5 shows the distribution of log quantities. In contrast to prices, here the litigated distribution is shifted *leftward*, reflecting the smaller quantities demanded in urgent purchases. The ordinary distribution has a heavier

Table A.6: Tender Success (LPM)

|                               | (1)                | (2)                | (3)                 | (4)                 |
|-------------------------------|--------------------|--------------------|---------------------|---------------------|
| <i>Panel A: Total Effect</i>  |                    |                    |                     |                     |
| Urgent Purchase               | 0.023**<br>(0.009) | 0.023**<br>(0.009) | 0.021***<br>(0.006) | 0.021***<br>(0.006) |
| <i>Panel B: Direct Effect</i> |                    |                    |                     |                     |
| Urgent Purchase               | 0.023**<br>(0.009) | 0.023**<br>(0.010) | 0.021***<br>(0.006) | 0.022***<br>(0.006) |
| Log Quantity                  | 0.003<br>(0.004)   | 0.003<br>(0.004)   | 0.012***<br>(0.002) | 0.012***<br>(0.002) |
| Item FE                       | Yes                | Yes                | Yes                 | Yes                 |
| Year FE                       | No                 | Yes                | Yes                 | No                  |
| Year-Month FE                 | No                 | No                 | No                  | Yes                 |
| PBU FE                        | No                 | No                 | Yes                 | Yes                 |
| Obs. (Panel A)                | 226,305            | 226,305            | 226,301             | 226,301             |
| Obs. (Panel B)                | 226,282            | 226,282            | 226,278             | 226,278             |
| Within R <sup>2</sup> (A)     | 0.001              | 0.001              | 0.000               | 0.000               |
| Within R <sup>2</sup> (B)     | 0.001              | 0.001              | 0.004               | 0.004               |

*Notes:* Dependent variable: successful tender (LPM). Sample: all observations. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses.  
\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

right tail, consistent with the ability to aggregate demand under standard planning timelines. The administrative distribution is bimodal, spanning both small (individual prescription) and larger (stock replenishment) quantities.

Figure A.6 displays the distribution of log number of bidding firms. The litigated distribution is clearly shifted leftward relative to ordinary purchases, with a larger mass at low participation levels. This visual pattern confirms the regression finding that urgent tenders attract fewer bidding firms, reducing the competitive pressure that normally restrains prices.

Figure A.7 focuses on the “under the gun” subsample, comparing the

Table A.7: Falsification: Items Never Subject to Litigation

|                       | <i>DV: log negotiated price</i> |                       | <i>DV: log reference price</i> |                      |
|-----------------------|---------------------------------|-----------------------|--------------------------------|----------------------|
|                       | (1)                             | (2)                   | (3)                            | (4)                  |
| Urgent Purchase       | −0.0204<br>(0.0316)             | 0.0513***<br>(0.0153) | −0.0306<br>(0.0447)            | 0.0305**<br>(0.0141) |
| Observations          | 39,283                          | 196,883               | 46,874                         | 226,278              |
| R <sup>2</sup>        | 0.834                           | 0.868                 | 0.811                          | 0.852                |
| Within R <sup>2</sup> | $7.0 \times 10^{-6}$            | 0.00022               | $1.7 \times 10^{-5}$           | $7.4 \times 10^{-5}$ |
| Sample                | Never-litigated                 | Full                  | Never-litigated                | Full                 |
| Item FE               | ✓                               | ✓                     | ✓                              | ✓                    |
| Year FE               | ✓                               | ✓                     | ✓                              | ✓                    |
| PBU FE                | ✓                               | ✓                     | ✓                              | ✓                    |

*Notes:* Columns (1) and (3) restrict the sample to items with zero litigated purchases across 2009–2019, keeping only those with both ordinary and administrative purchases for within-item comparability. Columns (2) and (4) reproduce the full-sample estimates from Tables ?? and A.3 for reference. The urgency coefficient is economically small and statistically indistinguishable from zero in the never-litigated sample, consistent with the urgency premium arising from exposure to judicial sanctions rather than from generic features of urgency. Standard errors clustered at the PBU level. Significance: \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

distributions of log negotiated prices for administrative and litigated urgent purchases only. The rightward shift of the litigated distribution relative to administrative purchases provides direct visual evidence of the sanction channel: holding constant the urgency of the purchase, judicial pressure shifts the entire price distribution to the right.

Figure A.8 compares tender success rates across purchase types. Consistent with the LPM estimates, litigated purchases have the highest success rate, followed by administrative and ordinary purchases. This ordering is consistent with the “under the gun” mechanism: the greater the penalty for procurement failure, the higher the observed success rate, as officials accept less favorable terms to avoid sanctions.



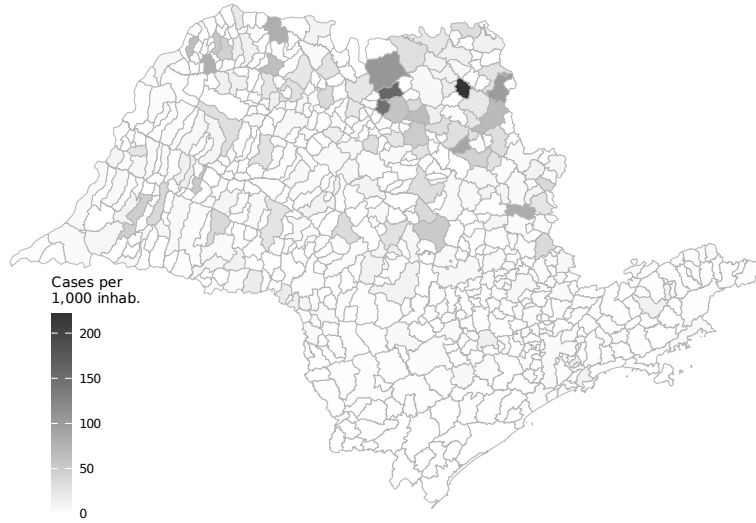


Figure A.1: Health Litigation Cases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

*Source:* Authors’ elaboration based on CNJ/INSPER (2019) litigation data and IBGE population estimates.

Figure A.9 plots mean log negotiated prices over time for ordinary and urgent purchases separately. The figure reveals two patterns. First, the price gap between urgent and ordinary purchases is persistent throughout the sample period, ruling out the possibility that the regression estimates are driven by a specific subperiod. Second, both series exhibit common trends, supporting the identification assumption that time-varying confounders do not differentially affect the two purchase types.

#### *A.4 Administrative Demand: Descriptive Evidence*

This section presents geographic descriptive evidence on the distribution of administrative demand across municipalities in São Paulo, complementing the litigation maps in Section ?? . Administrative purchases—urgent procurements initiated through internal government requests rather than court



Figure A.2: Public Buyer Units (SES/SP)

*Source:* Authors' elaboration based on SES/SP administrative records.

orders—share the same planning constraints as litigated purchases but carry no sanctions for procurement failure. The spatial patterns documented below confirm that administrative and litigated demand originate from similar geographic areas, supporting the institutional comparability that underlies the “under the gun” identification strategy.

Figure A.10 maps administrative purchases per 1,000 inhabitants across municipalities. The geographic distribution closely mirrors the litigation pattern shown in Figure A.1, with higher per capita rates in municipalities that host PBUs and in the interior of the state. This spatial overlap is consistent with both channels responding to the same underlying health needs.

Figure A.11 shows the location of PBUs that process administrative purchases, with marker size proportional to the volume of transactions. The distribution mirrors the PBU map for litigated purchases (Figure A.2), confirming that the same institutional infrastructure handles both procurement

| Table A.8: Robustness: Reference Prices |                     |                     |                   |                    |
|-----------------------------------------|---------------------|---------------------|-------------------|--------------------|
|                                         | (1)                 | (2)                 | (3)               | (4)                |
| <i>Panel A: No Winsorization</i>        |                     |                     |                   |                    |
| Urgent Purchase                         | 0.167***<br>(0.056) | 0.159***<br>(0.051) | 0.026*<br>(0.014) | 0.026*<br>(0.014)  |
| <i>Panel B: Winsorized 1%/99%</i>       |                     |                     |                   |                    |
| Urgent Purchase                         | 0.164***<br>(0.056) | 0.156***<br>(0.050) | 0.027*<br>(0.014) | 0.027*<br>(0.014)  |
| <i>Panel C: Winsorized 5%/95%</i>       |                     |                     |                   |                    |
| Urgent Purchase                         | 0.156***<br>(0.053) | 0.150***<br>(0.048) | 0.027*<br>(0.014) | 0.028**<br>(0.014) |
| Item FE                                 | Yes                 | Yes                 | Yes               | Yes                |
| Year FE                                 | No                  | Yes                 | Yes               | No                 |
| Year-Month FE                           | No                  | No                  | No                | Yes                |
| PBU FE                                  | No                  | No                  | Yes               | Yes                |
| Observations                            | 196,899             | 196,899             | 196,896           | 196,896            |

*Notes:* Dependent variable: log reference price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

channels.

Figure A.12 provides a visual comparison of administrative and litigated purchases, highlighting that the procurement process is identical across both channels—the only distinguishing feature is the threat of judicial sanctions in litigated cases. This institutional parallel is the foundation for the “under the gun” identification.

Figure A.13 maps the total number of administrative purchases by municipality. As with litigation, administrative demand is concentrated in a small number of urban centers, particularly the São Paulo metropolitan area and regional hubs in the state’s interior.

Figure A.14 shows the share of administrative purchases as a proportion of total purchases (administrative plus litigated) by municipality. The sub-

Table A.9: Robustness: Negotiated Prices

|                                   | (1)                 | (2)                 | (3)                 | (4)                 |
|-----------------------------------|---------------------|---------------------|---------------------|---------------------|
| <i>Panel A: No Winsorization</i>  |                     |                     |                     |                     |
| Urgent Purchase                   | 0.160***<br>(0.043) | 0.152***<br>(0.038) | 0.051***<br>(0.015) | 0.054***<br>(0.016) |
| <i>Panel B: Winsorized 1%/99%</i> |                     |                     |                     |                     |
| Urgent Purchase                   | 0.159***<br>(0.042) | 0.151***<br>(0.038) | 0.053***<br>(0.016) | 0.056***<br>(0.016) |
| <i>Panel C: Winsorized 5%/95%</i> |                     |                     |                     |                     |
| Urgent Purchase                   | 0.162***<br>(0.041) | 0.153***<br>(0.037) | 0.058***<br>(0.017) | 0.061***<br>(0.017) |
| Item FE                           | Yes                 | Yes                 | Yes                 | Yes                 |
| Year FE                           | No                  | Yes                 | Yes                 | No                  |
| Year-Month FE                     | No                  | No                  | No                  | Yes                 |
| PBU FE                            | No                  | No                  | Yes                 | Yes                 |
| Observations                      | 196,886             | 196,886             | 196,883             | 196,883             |

*Notes:* Dependent variable: log negotiated price. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

stantial variation across municipalities suggests that the relative importance of administrative versus litigated demand reflects local institutional factors—such as the presence of administrative request mechanisms and the propensity of local courts to grant injunctions—rather than a uniform statewide pattern.

#### A.5 Event Study: First Court Order (*Honest DiD*)

To address concerns about pre-trends and heterogeneous-treatment-effect contamination in the TWFE event-study design, we re-estimate the dynamic response of log negotiated prices to the first court order for each item using two contemporary estimators: the imputation estimator of ? ] (BJS) and the  $ATT(g, t)$  estimator of ? ] (CS). The panel is at the item-year level with winner-only observations; treatment is defined as the calendar year of the first litigated purchase for each item; the BJS specification uses item + year

Table A.10: Robustness: Participant Firms

|                                   | (1)                  | (2)                  | (3)                  | (4)                  |
|-----------------------------------|----------------------|----------------------|----------------------|----------------------|
| <i>Panel A: No Winsorization</i>  |                      |                      |                      |                      |
| Urgent Purchase                   | -0.101***<br>(0.023) | -0.113***<br>(0.022) | -0.056***<br>(0.015) | -0.054***<br>(0.015) |
| <i>Panel B: Winsorized 1%/99%</i> |                      |                      |                      |                      |
| Urgent Purchase                   | -0.101***<br>(0.023) | -0.113***<br>(0.022) | -0.056***<br>(0.014) | -0.054***<br>(0.015) |
| <i>Panel C: Winsorized 5%/95%</i> |                      |                      |                      |                      |
| Urgent Purchase                   | -0.100***<br>(0.022) | -0.111***<br>(0.021) | -0.055***<br>(0.014) | -0.053***<br>(0.015) |
| Item FE                           | Yes                  | Yes                  | Yes                  | Yes                  |
| Year FE                           | No                   | Yes                  | Yes                  | No                   |
| Year-Month FE                     | No                   | No                   | No                   | Yes                  |
| PBU FE                            | No                   | No                   | Yes                  | Yes                  |
| Obs. (A)                          | 81,377               | 81,377               | 81,373               | 81,373               |
| Obs. (B)                          | 81,805               | 81,805               | 81,801               | 81,801               |
| Obs. (C)                          | 81,805               | 81,805               | 81,801               | 81,801               |

*Notes:* Dependent variable: log number of bidding firms. Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

fixed effects fit on untreated observations to impute the counterfactual. The CS specification uses never-litigated items as the control group.

BJS is our preferred honest specification because it uses pre-treatment observations of eventually-treated items to fit the counterfactual model, avoiding the selection problem that haunts the CS nevertreated comparison. The pattern is consistent with the cross-sectional estimates in the main text: after a court order, the same item is systematically more expensive than it was in its own pre-period, with no detectable pre-trend that could confound the interpretation.

Table A.11: Robustness: Tender Success (LPM)

|                                   | (1)                | (2)                | (3)                 | (4)                 |
|-----------------------------------|--------------------|--------------------|---------------------|---------------------|
| <i>Panel A: No Winsorization</i>  |                    |                    |                     |                     |
| Urgent Purchase                   | 0.023**<br>(0.009) | 0.023**<br>(0.009) | 0.021***<br>(0.006) | 0.021***<br>(0.006) |
| <i>Panel B: Winsorized 1%/99%</i> |                    |                    |                     |                     |
| Urgent Purchase                   | 0.023**<br>(0.009) | 0.023**<br>(0.009) | 0.021***<br>(0.006) | 0.021***<br>(0.006) |
| <i>Panel C: Winsorized 5%/95%</i> |                    |                    |                     |                     |
| Urgent Purchase                   | 0.023**<br>(0.009) | 0.023**<br>(0.009) | 0.021***<br>(0.006) | 0.021***<br>(0.006) |
| Item FE                           | Yes                | Yes                | Yes                 | Yes                 |
| Year FE                           | No                 | Yes                | Yes                 | No                  |
| Year-Month FE                     | No                 | No                 | No                  | Yes                 |
| PBU FE                            | No                 | No                 | Yes                 | Yes                 |
| Observations                      | 226,305            | 226,305            | 226,301             | 226,301             |

*Notes:* Dependent variable: successful tender (LPM). Sample: winners only (success uses all obs). Each panel uses a different winsorization level. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

### A.6 Heterogeneity

We examine heterogeneity in the urgency premium along four dimensions. Tables A.13–A.16 report results from split-sample regressions and interaction models using our preferred specification (item + year + PBU FE).

*SUS component (Table A.13).* The urgency premium on negotiated prices is concentrated among items in the basic SUS component (common medications): the coefficient is 0.030 ( $p < 0.10$ ) for basic items and an insignificant 0.005 for specialized items, suggesting that procurement disruption is more consequential for items where established supply chains normally deliver better prices.

Table A.12: Robustness: Under the Gun with Progressive Controls

|                                   | (1)                 | (2)              | (3)                | (4)                | (5)                |
|-----------------------------------|---------------------|------------------|--------------------|--------------------|--------------------|
| <i>Panel A: No Winsorization</i>  |                     |                  |                    |                    |                    |
| Administrative (vs Litigated)     | -0.259**<br>(0.095) | 0.138<br>(0.130) | 0.115**<br>(0.054) | 0.201**<br>(0.090) | 0.195**<br>(0.087) |
| <i>Panel B: Winsorized 1%/99%</i> |                     |                  |                    |                    |                    |
| Administrative (vs Litigated)     | -0.262**<br>(0.096) | 0.117<br>(0.122) | 0.110*<br>(0.055)  | 0.199**<br>(0.093) | 0.194**<br>(0.090) |
| <i>Panel C: Winsorized 5%/95%</i> |                     |                  |                    |                    |                    |
| Administrative (vs Litigated)     | -0.261**<br>(0.096) | 0.060<br>(0.093) | 0.089*<br>(0.049)  | 0.169*<br>(0.084)  | 0.165*<br>(0.080)  |
| Log Quantity                      | No                  | Yes              | Yes                | Yes                | Yes                |
| Log Ref. Price                    | No                  | No               | Yes                | Yes                | Yes                |
| Log N. Firms                      | No                  | No               | No                 | Yes                | Yes                |
| Item FE                           | Yes                 | Yes              | Yes                | Yes                | Yes                |
| Year FE                           | Yes                 | Yes              | Yes                | Yes                | No                 |
| Year-Month FE                     | No                  | No               | No                 | No                 | Yes                |
| PBU FE                            | Yes                 | Yes              | Yes                | Yes                | Yes                |
| Obs. (A)                          | 56,803              | 56,803           | 56,803             | 14,913             | 14,913             |
| Obs. (B)                          | 56,803              | 56,803           | 56,803             | 15,052             | 15,052             |
| Obs. (C)                          | 56,803              | 56,803           | 56,803             | 15,052             | 15,052             |

*Notes:* Dependent variable: log negotiated price. UTG sample (urgent, winners, items with both administrative and litigated). Progressive controls added left to right. Column (5) replaces Year with Year-Month FE. Standard errors clustered at the PBU level in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

*Time period (Table A.14).* The urgency premium is larger in the later period (2014–2019) than in the earlier period (2009–2013), consistent with the growing volume and complexity of health litigation over time.

*Market competition (Table A.15).* The urgency premium is three times larger in competitive markets (0.143 vs. 0.046), with a significant interaction (0.226,  $p < 0.01$ ). In concentrated markets, fewer bidders leave less room for urgency to erode bargaining power; in competitive markets, the loss of planning time has the largest marginal impact.

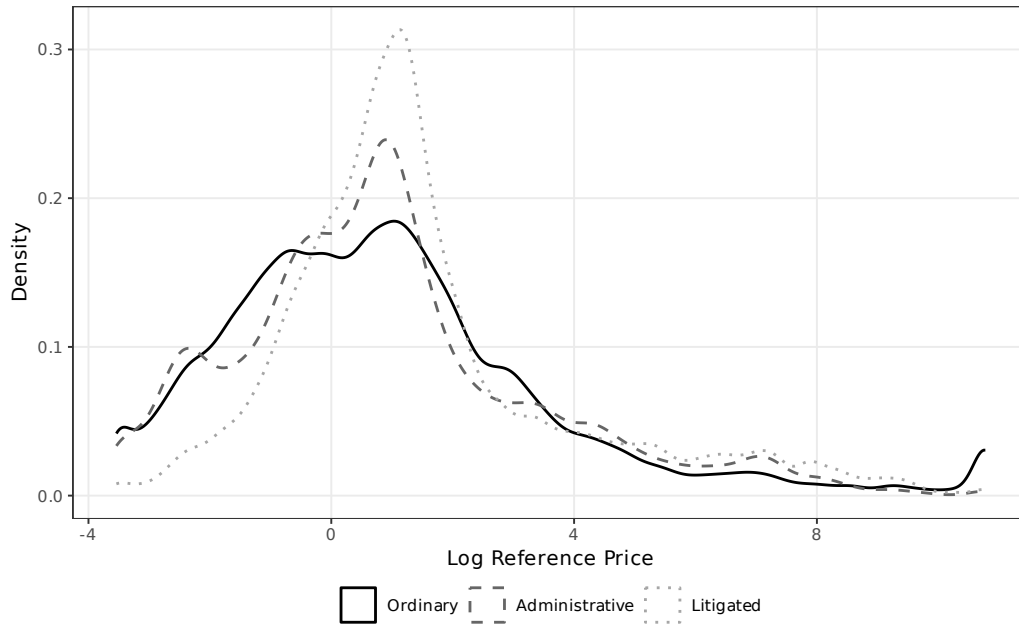


Figure A.3: Distribution of Log Reference Price by Purchase Type

*PBU size* (Table A.16).. The urgency premium is slightly larger for purchases made by smaller PBUs (below-median transaction volume), consistent with smaller units having less experience managing urgent procurement.



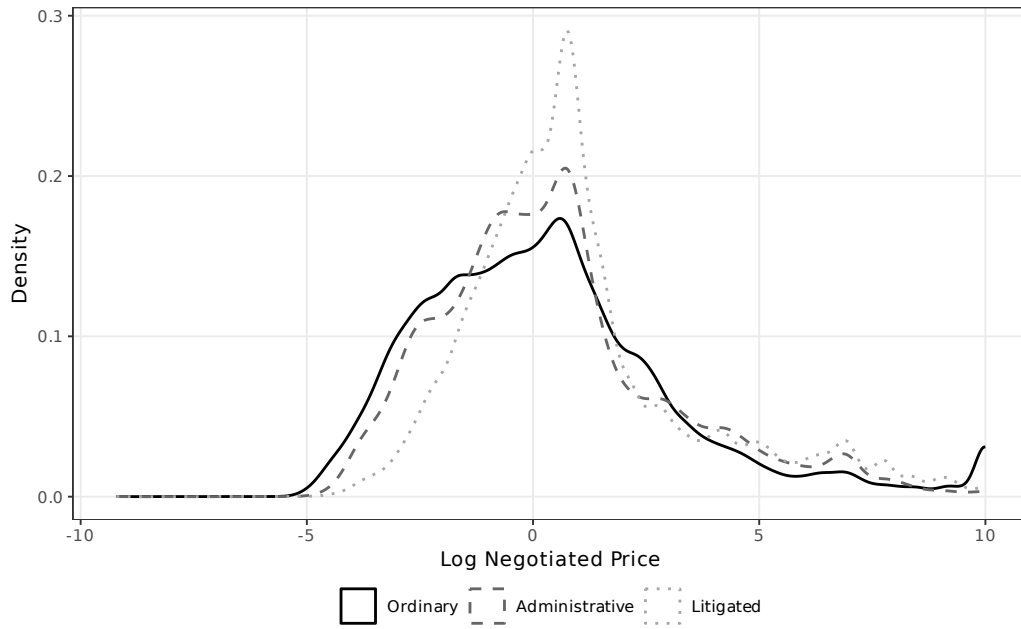


Figure A.4: Distribution of Log Negotiated Price by Purchase Type

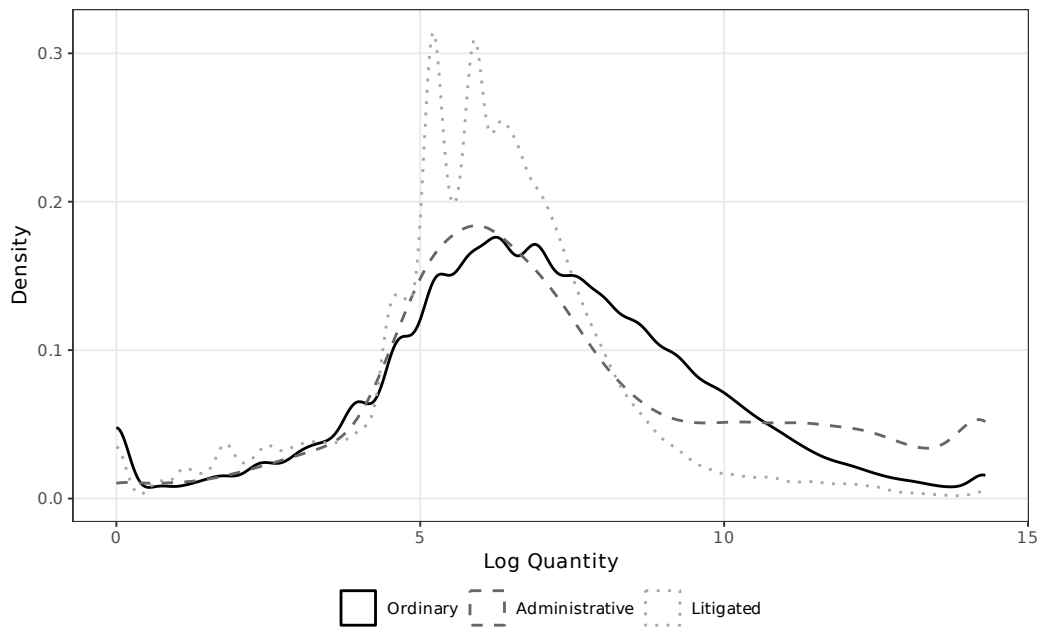


Figure A.5: Distribution of Log Quantity by Purchase Type

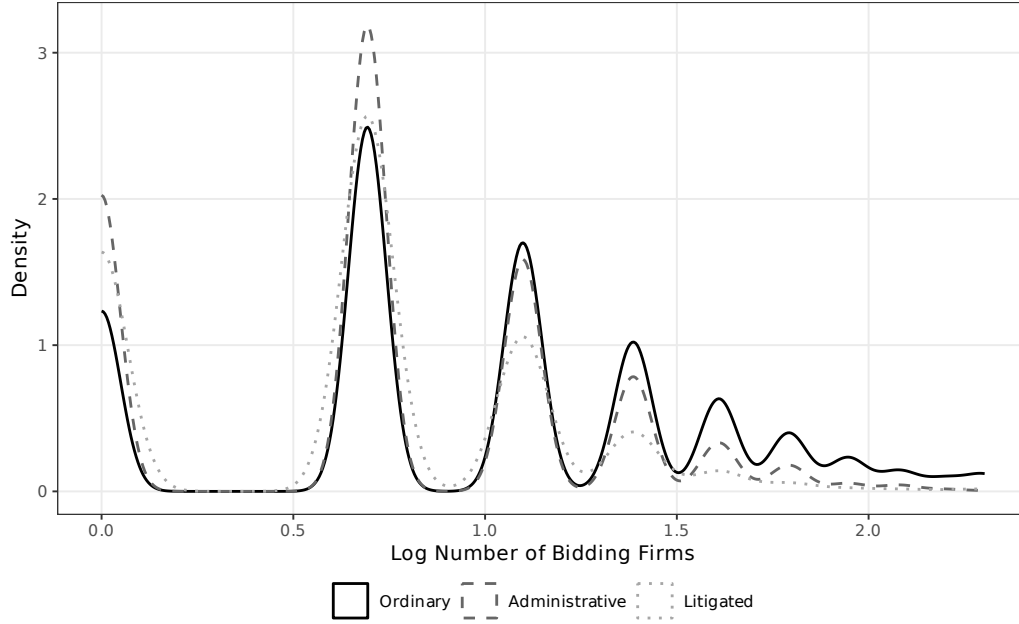


Figure A.6: Distribution of Log Number of Bidding Firms by Purchase Type

Table A.13: Heterogeneity: SUS Component

|                           | Split Sample       |                   | Interaction        |
|---------------------------|--------------------|-------------------|--------------------|
|                           | Specialized<br>(1) | Basic SUS<br>(2)  | Full Sample<br>(3) |
| Urgent Purchase           | 0.005<br>(0.046)   | 0.030*<br>(0.016) | -0.073<br>(0.097)  |
| Urgent $\times$ Basic SUS |                    |                   | 0.159<br>(0.115)   |
| FE: Item + Year + PBU     | Yes                | Yes               | Yes                |
| Observations              | 45,048             | 151,835           | 196,883            |
| Within R <sup>2</sup>     | 0.000              | 0.000             | 0.001              |

*Notes:* Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

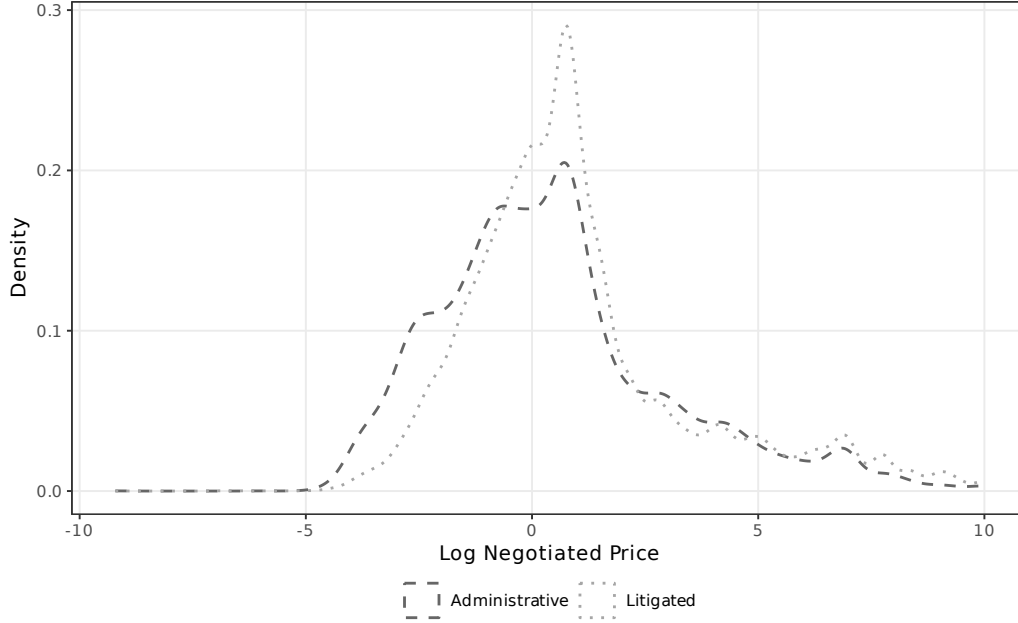


Figure A.7: Distribution of Log Negotiated Price: Administrative vs. Litigated (Urgent Only)

Table A.14: Heterogeneity: Time Period

|                             | Split Sample        |                     | Interaction          |
|-----------------------------|---------------------|---------------------|----------------------|
|                             | Early<br>(1)        | Late (2014+)<br>(2) | Full Sample<br>(3)   |
| Urgent Purchase             | 0.074***<br>(0.025) | 0.045**<br>(0.018)  | 0.259***<br>(0.035)  |
| Urgent $\times$ Late Period |                     |                     | -0.339***<br>(0.043) |
| FE: Item + Year + PBU       | Yes                 | Yes                 | Yes                  |
| Observations                | 82,658              | 113,936             | 196,883              |
| Within R <sup>2</sup>       | 0.000               | 0.000               | 0.005                |

*Notes:* Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

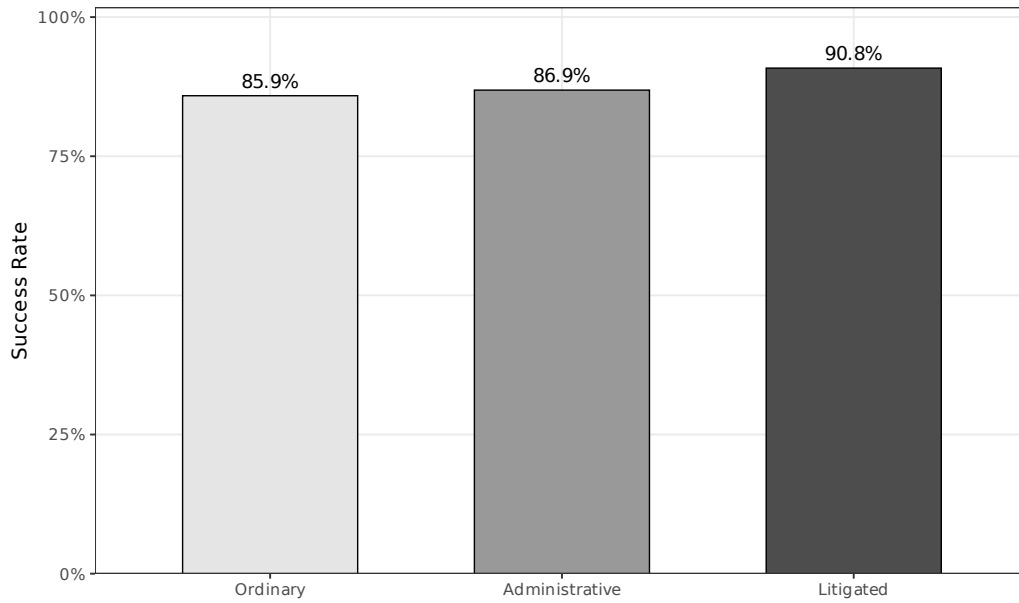


Figure A.8: Tender Success Rate by Purchase Type

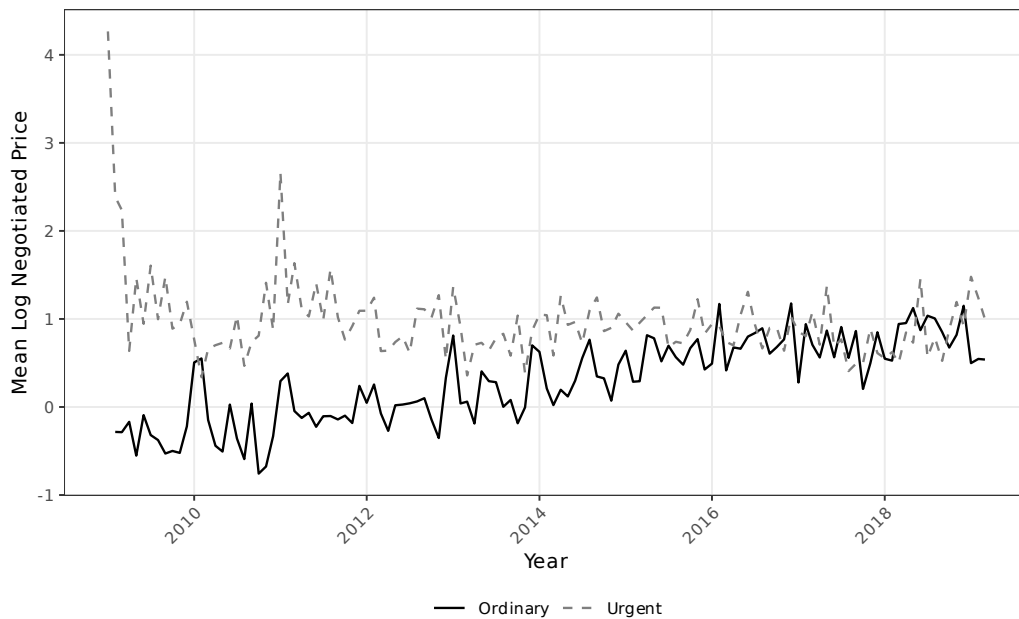


Figure A.9: Mean Log Negotiated Price Over Time: Ordinary vs. Urgent



Figure A.10: Administrative Purchases per 1,000 Inhabitants Across Municipalities in São Paulo (2009–2019)

*Source:* Authors' elaboration based on BEC procurement data and IBGE population estimates.

Table A.15: Heterogeneity: Market Competition

|                                  | Split Sample        |                     | Interaction         |
|----------------------------------|---------------------|---------------------|---------------------|
|                                  | Low Comp.<br>(1)    | High Comp.<br>(2)   | Full Sample<br>(3)  |
| Urgent Purchase                  | 0.046***<br>(0.016) | 0.143***<br>(0.034) | 0.026<br>(0.018)    |
| Urgent $\times$ High Competition |                     |                     | 0.226***<br>(0.044) |
| FE: Item + Year + PBU            | Yes                 | Yes                 | Yes                 |
| Observations                     | 125,950             | 61,313              | 187,264             |
| Within R <sup>2</sup>            | 0.000               | 0.001               | 0.001               |

*Notes:* Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

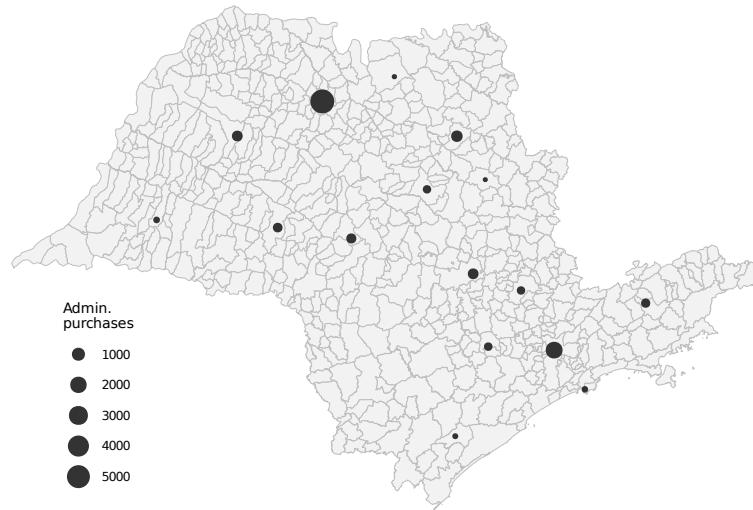


Figure A.11: PBUs Handling Administrative Purchases (Size  $\propto$  Volume)  
*Source:* Authors' elaboration based on SES/SP administrative records.

|                             | Administrative  | Litigated            |
|-----------------------------|-----------------|----------------------|
| <b>Origin</b>               | SES/SP request  | Court order          |
| <b>Source of funds</b>      | Diverted budget | Diverted budget      |
| <b>Quantity</b>             | Small           | Small                |
| <b>Delivery time</b>        | Short           | Short                |
| <b>Threat of punishment</b> | None            | Potential punishment |

Figure A.12: Comparison of Administrative and Litigated Purchases  
*Source:* Authors' elaboration.

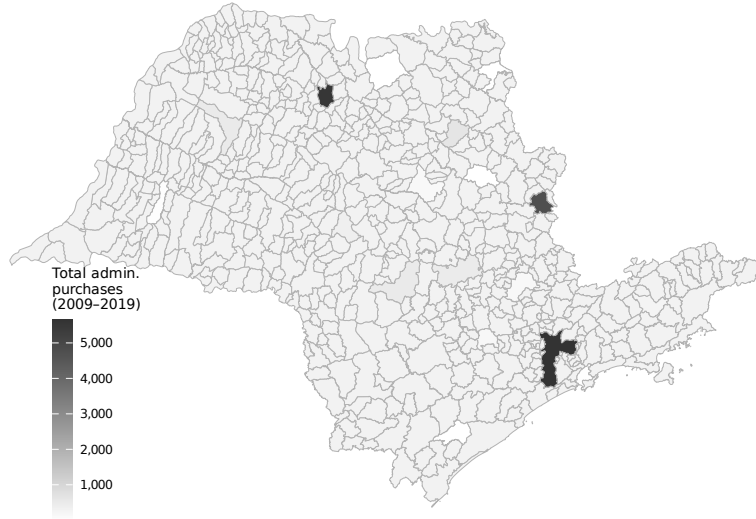


Figure A.13: Total Administrative Purchases by Municipality in São Paulo (2009–2019)

*Source:* Authors' elaboration based on BEC procurement data.

Table A.16: Heterogeneity: PBU Size

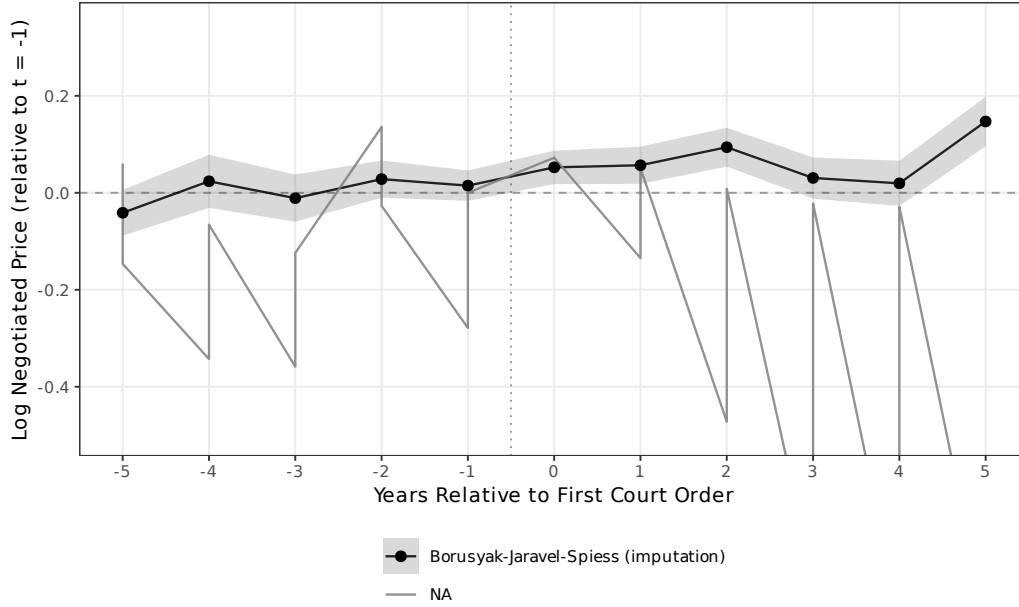
|                           | Split Sample     |                     | Interaction        |
|---------------------------|------------------|---------------------|--------------------|
|                           | Small PBU<br>(1) | Large PBU<br>(2)    | Full Sample<br>(3) |
| Urgent Purchase           | 0.023<br>(0.026) | 0.062***<br>(0.018) | -0.000<br>(0.049)  |
| Urgent $\times$ Large PBU |                  |                     | 0.070<br>(0.054)   |
| FE: Item + Year + PBU     | Yes              | Yes                 | Yes                |
| Observations              | 31,902           | 164,511             | 196,883            |
| Within R <sup>2</sup>     | 0.000            | 0.000               | 0.000              |

*Notes:* Preferred specification: Item + Year + PBU FE; PBU-clustered SEs in parentheses. Columns (1)–(2) are split-sample estimates. Column (3) reports the interaction. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .



Figure A.14: Administrative Purchases as Share of Total Urgent Purchases by Municipality  
*Source:* Authors' elaboration based on BEC procurement data.





and 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.15: Log Negotiated Price Around First Court Order: Honest DiD Estimators

*Notes:* Item-year panel, winner-only observations. BJS ( ) shown with shaded 95% confidence interval; CS ( ) with never-treated control group; TWFE shown only as a pre-reform benchmark. The BJS pre-period coefficients are economically small (all  $< 0.05$  in absolute value) and statistically indistinguishable from zero, consistent with parallel trends; post-treatment coefficients rise monotonically from  $+0.052$  at  $t = 0$  to  $+0.147$  at  $t = +5$ . The CS estimator with never-treated control is substantially noisier because items that never attract a court order are a systematically different comparison group (they tend to be routine, well-stocked items); HonestDiD ( -compatible sensitivity in the companion CSV) confirms that the CS point estimate at  $t = 0$  is not robust to parallel-trend violations of moderate magnitude.